

Business Process Monitoring with S-BPM

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Abstract. Business Process Monitoring plays an important role in organizational development. It delivers management information and thus allows for better decision making in terms of process design, implementation and execution. This contribution first sheds some light on the importance and forms of business process monitoring. Then we look at current monitoring features of S-BPM software supporting the S-BPM approach and discuss development perspectives particularly in the field of Business Activity Monitoring (BAM). In this context we adapt an existing concept for modeling BAM specifications with reference to BPMN models for the use with S-BPM.

Keywords: Business Process Monitoring, Business Activity Monitoring (BAM), Complex Event Processing (CEP), Process Controlling, Metrics, Process Performance Indicators (PPI), Subject-oriented Business Process Management (S-BPM), BPMN.

1 Introduction

Monitoring of business processes looks at running instances. For those it measures metrics, aggregates them to Process Performance Indicators (PPIs) as a business process-related form of Key Performance Indicators (KPIs), reveals deviations (as-is vs. to-be) and reports and presents results to people in charge or interested in the value of the PPI. Thus monitoring lays ground for the performance analysis in the key dimensions quality, time and costs of processes and helps identifying weaknesses and opportunities for improvement [13].

By feeding back information for completed and running instances to analysis monitoring fosters organizational learning, forms an important part of the Business Process Management (BPM) lifecycle [26] and thus helps implementing the operational level in the closed-loop approach to enterprise performance management [27] (see figure 1).

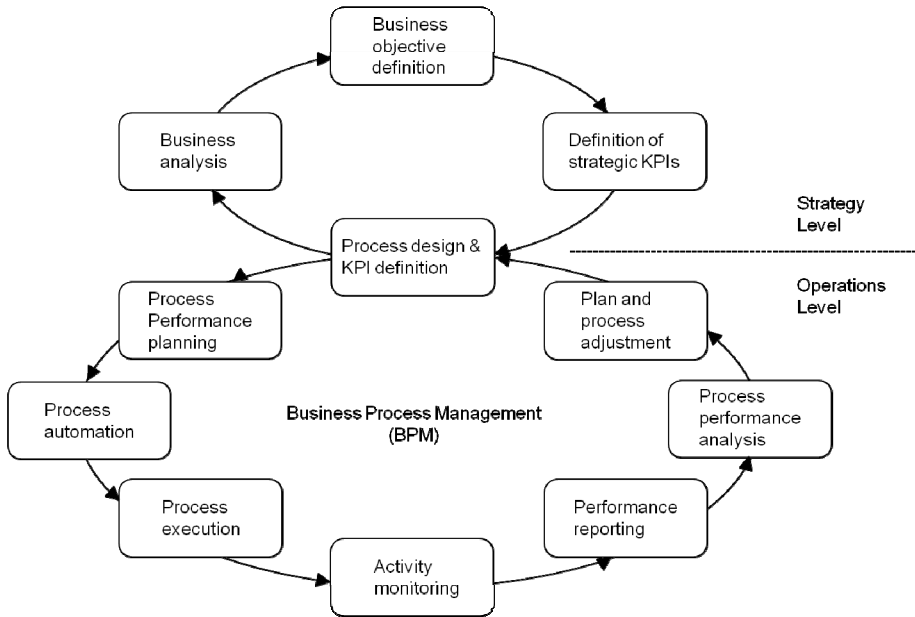


Fig. 1. Closed-loop Approach to Performance Management [6]

After this introduction we differentiate facets of Business Process Monitoring and briefly describe the current status and development perspectives of monitoring in the S-BPM context. Chapter 3 deals with Business Activity Monitoring for S-BPM, focusing on modeling BAM parameters. The contribution ends with a conclusion and a prospect on future work in chapter 4.

2 Business Process Monitoring

2.1 Facets

Business Process Monitoring as described above can be distinguished in an ex-post view and a real-time or near real-time view (see figure 2) [27]. The first one, also referred to as (operational) process controlling [31, 15], aims at collecting instance data from transactional systems and event logs, storing it on an aggregated level in a data warehouse and analyzing it on request or periodically. This traditional understanding of process monitoring (time- and request driven) is being complemented by Business Activity Monitoring (BAM). BAM is driven by events and uses Complex Event Processing (CEP) concepts (see section 3.1) to analyze instance data as it incurs, which means in real-time or near real-time [3, 7, 19]. A BAM/CEP environment aggregates single low-level events without meaning to complex ones and thus delivers timely insights into business processes. It can set alerts or initiate specified escalation procedures like instantiating a process or altering a running instance [27, 28, 30, 16, 19].

Under the term of Business Process Analytics Janiesch et al. add two more facets of monitoring and analysis [15]. The first is Process Mining which, as a special ex-post analysis, investigates event logs in order to identify instance behavior at runtime and derive models from it [29, 1]. The second one is Process Intelligence which intends to predict the future behavior of instances by simulating different settings (resources, model variants) using what-if, how-to-achieve and probabilistic scenarios (see also section 2.3).

Activities	Process Monitoring		
	Traditional Monitoring	← complements	Business Activity Monitoring
Measurement/Computing Objects	Metrics, performance indicators and other data from heterogenous sources (transactional systems, event logs)		
Analysis	Trigger & point in time Concepts & Methods Request (pull) Ex post Business Intelligence Store and analyze Classic Database Requests Data Warehouse/OLAP/Data Mining	Time (push) Operational Intelligence	Event (push) Real-time/near real-time (low latency) Complex Event Processing Stream and analyze Continuous Database Requests Stream Mining
Reporting & Presentation	Ad hoc	Periodically	Permanently (very short refreshing intervals) & by exception
Addressees	upper & top management, process owners, business analysts		operational management process owners & participants
Cause Analysis, Decision, Action	Usually mid-term/long-term		Immediately/short-term

Fig. 2. Facets of Business Process Monitoring [27]

2.2 State of the Art in Software Supporting Subject-Oriented Monitoring

Software currently available for running process instances according to S-BPM models already provides some monitoring functionality. Metasonic Flow as the process engine of the Metasonic Suite creates process instances and controls their execution at runtime. It involves users and applications or IT services as subject representatives and controls the handling of business objects as specified in the subject behavior. While doing this the engine collects all sorts of execution data, mainly by counting instances and setting time stamps for events like state transitions (e.g. begin time and end time of every single instance or steps within an instance). Using this information stored in its event log (audit log) the process engine can generate predefined reports on instance and process level including for example:

- Duration of running and completed instances, together with the duration status displayed as traffic light colors, depending on deviation of a maximum value specified in the process model (see figure 3).

Process name ▼ ▲	Priority ▼ ▲	Title ▼ ▲	Creator ▼ ▲	Start time ▼ ▲	Running time	Duration state
Ordering (Simple Version)		My Metasonic Proof Process # 1		30.06.2011	0 day(s)	
Supply (Extended Version)	Normal	06/15/2011 11:54 - Ordering (Extended)_JoDo+	John Doe	15.06.2011	15 day(s)	
Ordering (Extended Version)	Normal	06/15/2011 11:54 - Ordering (Extended)_JoDo+	John Doe	15.06.2011	15 day(s)	
Ordering (Simple Version)	High	06/15/2011 11:48 - Ordering (Simple)_John_Clipbo...	John Doe	15.06.2011	0 day(s)	
Team Building	Normal	06/15/2011 11:43 - Team Building_MatHu	Max Mustermann	15.06.2011	15 day(s)	
Ordering (Extended Version)	Normal	06/15/2011 11:42 - Ordering (Extended)_DiRi	Diane Richards	15.06.2011	15 day(s)	

Fig. 3. Instance Report 1 [21]

- Duration of a running instance (absolute value and comparison with average), chronological sequence and duration of the single steps performed by the involved subjects (see figure 4).

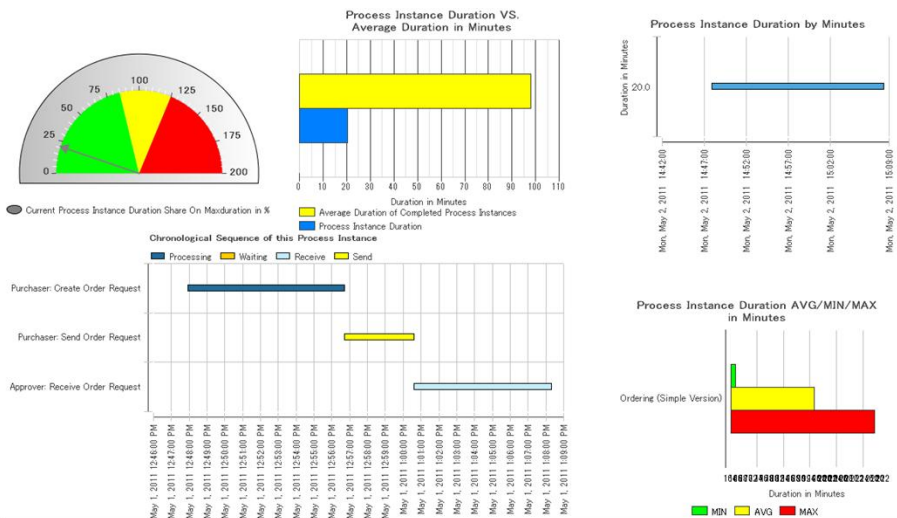


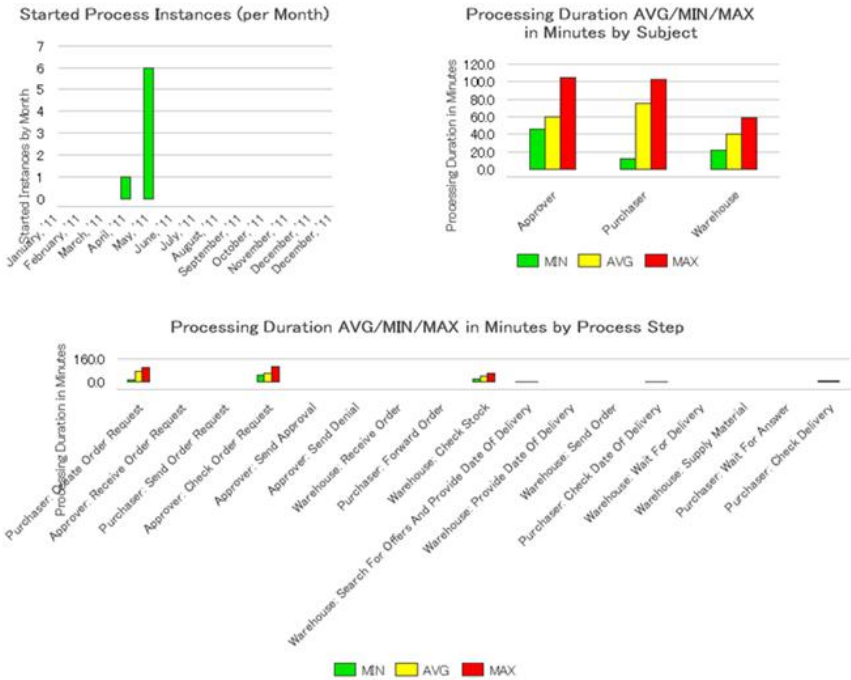
Fig. 4. Instance Report 2 [21]

- Sequence of process steps of a running instance with time stamps for the completion of steps, allowing to answer questions regarding the current processing state (how far has my order been processed already? which subject needs to perform which activity next?) (see figure 5).

Purchaser	Approver	Warehouse
Create Order Request State Type: Function State left: Fri May 06 10:07:39 CEST 2011 Editor:John Doe Title open Flipchart for John		
Send Order Request State Type: Send State left: Fri May 06 10:07:51 CEST 2011 to Approver, Receive Order Request Editor:John Doe	Receive Order Request State Type: Receive State left: Fri May 06 10:07:51 CEST 2011 Receiving Purchaser, Send Order Request Editor:Refinement	

Fig. 5. Instance Report 3 [21]

- Number of instances of a process created per period (e.g. month), average, minimum and maximum processing time per subject and per process step (activity) (see figure 6).



ID	PROCESS INSTANCE NAME	START	END		VAL MIN.	AVG MIN.	% AVG	
1	21/04/2011 15:29 - Ordering (Extended) Ve04/21/11 15:29:46	05/03/11 15:56:20			17306.57	359.69	4811.6%	
7	05/02/2011 11:09 - Ordering (Extended)1	05/02/11 11:09:32	05/02/11 14:30:57		201.42	359.69	56.0%	
8	05/02/2011 11:09 - Ordering (Extended)2	05/02/11 11:09:45	05/02/11 14:47:36		217.85	359.69	60.6%	
9	05/02/2011 11:09 - Ordering (Extended)3	05/02/11 11:10:03	05/02/11 17:23:34		373.52	359.69	103.8%	

Fig. 6. Process Report [21]

2.3 Development Perspectives in the S-BPM Context

Activity-Based Costing (ABC). Due to explicitly considering the subjects cost information can be relatively easy integrated into S-BPM. Embedding the subjects into the organization, which, at the end, means assigning them to people, allows determining personnel-related costs. Measuring the processing time subjects spend on their activities and multiplying it by the wage rates for the real people representing the subjects leads to realistic cost values for a process. As people belong to organizational units and cost centers usually are assigned to these, it is possible to compute process costs at cost center level. For more details see [31, 32].

Simulation. Simulating a subject-oriented process model allows investigating the performance of the process with varying resource assignments (especially personnel) as well as with variants of the process structure. A concept and a software tool supporting simulation in S-BPM is currently being developed (see [24] for details). [15] mention the use of simulation either before a process goes live or after it has run for while. We suggest to also applying it at runtime in the context of Business Activity Monitoring. A real-time simulation during processing of an instance could deliver decision support in case the execution should be altered. For example if monitoring signals a delay after 3 of 9 steps the question could be whether suspending other instances or increasing resources in later steps is possible without violating other constraints. Simulating a respective change and its consequences can help to take the right decision in the sense of a predictive workflow management [4, 22].

After briefly mentioning and referencing the S-BPM related activities on Activity-based Costing and Simulation we are going to look at Business Activity Monitoring, starting with an overview of a general BAM/CEP architecture and then concentrating on the modeling of BAM parameters at build time.

3 Business Activity Monitoring for S-BPM

3.1 Architecture

A BAM environment supported by Complex Event Processing consists of several elements necessary at build time and at runtime (see figure 7) [27, 8, 15]. At build time a modeling environment should provide tools for designing processes (e.g. Metasonic Build) and defining process performance indicators (PPIs), BAM events, rules, thresholds etc. as well as parameters for their visualization in reports and on dashboards. At runtime there are (1) event producers like a process engine (e.g. Metasonic Flow) or an ERP system (e.g. SAP) which feed events into an event cloud or stream (chronologically ordered). (2) Event Processing Agents (EPA) form the event processing logic. They process events based on metrics, event patterns, rules and other parameters specified at design time. Their basic logical functions include

filtering and transforming events and detecting patterns among them. Global state elements allow them accessing data from outside the application (e.g. from an ERP system). EPAs put the results of their processing (also to be understood as events) out to Event Consumers (3) like dashboards or process engines. Input and Output Adapters (IA, OA) transform event data between different formats of system elements as necessary. All system elements involved form an Event Processing Network (EPN), in which events are exchanged by communication mechanisms.

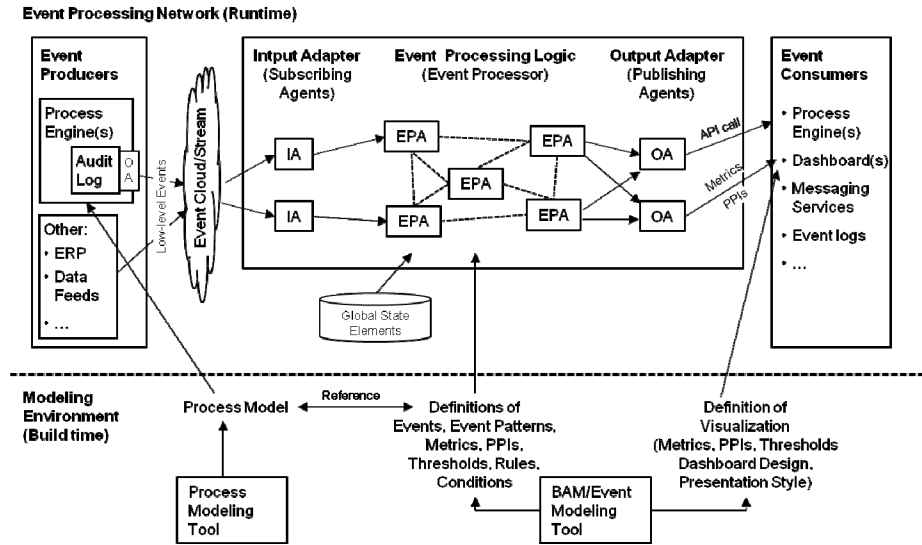


Fig. 7. Integrated BAM/CEP Architecture [27]

3.2 Modeling BAM Parameters at Build Time

As mentioned in the last section it is necessary not only to model the processes, but also numerous pieces of information relevant for a sound process monitoring in the sense of Business Activity Monitoring (BAM model). These can be derived from answers to questions like what, when, how and how often should be measured by whom [25]. The information should also include how single metrics are to be aggregated in order to determine Process Performance Indicators (PPIs). For systematically collecting and documenting the necessary information fact sheets or templates for metrics and performance indicators have been developed [17, 20]. Table 1 shows an extract of a sample fact sheet defined for the average processing time of activities (see also [31, 23]). Definitions like this are in parts basis for the reports presented in section 2.2. They can be considered to be a first model for describing BAM parameters.

Table 1. Fact Sheet for a PPI (extract)

Attribute	Content
Characteristics	
Identifier	Average activity time
Description	Average time of a process activity within a certain period
To-be value/unit	tbd specifically (min.)
Tolerance range/unit	tbd specifically (%)
Escalation Rules/ Actions	In case of violation alert the process owner and start escalation process (tbd specifically)
Addressees	Process Owner, Middle Mangement, Accountants (tbd specifically)
Responsibility	Process Owner (tbd specifically)
Measuring and Computing	
Measuring Object	All instances of the process 'Purchase Order'
(Single) Metrics	Start time and end time of all activities of the process
Measuring Method	Read time stamps for beginning and end of activities written by Metasonic Flow
Measuring Frequency	For every single instance as it occurs
Algorithms	For computing period: $\frac{\text{Activity processing time}}{\text{Completed instances}}$
Data Sources (general)	Tables in the database of Metasonic Suite: RT_PROCDesc, RT_PROCIInst, REC_PARADesc, REC_RECTrans, UM_USER
Data Sources (specific)	Activity processing time (for one instance): <pre> SELECT TIMESTAMP1 – (SELECT STARTTIME FROM RT_PROCIInst WHERE RT_PROCDesc = <i>process (purchase order)</i> AND ID = <i>instance (9)</i> FROM REC_RECTrans WHERE RT_STDesc = <i>state (fill_in_form)</i> AND RT_PROCIInst = <i>instance (9)</i> </pre> Completed instances: see separate fact sheet
Computing Period (time, no. of inst.)	Daily
Presentation	
Presentation Style	As-is value and to-be value in combination with a spark line showing the historical development, deviation from to-be value in %
Presentation Frequency	Weekly and in case of escalation
Archiving	Stored in additional database table, linked with RT_PROCDesc

Replacing the content column by more formal ontology-based linguistic patterns as suggested by Del-Río-Ortega et al. (see table 2) could help relating PPIs to elements of the process model, performing automated analysis [5] and implementing the measurement at runtime.

Table 2. PPI Template based on Linguistic Patterns [5]

Attribute	Linguistic Pattern	Example
PPI-<ID>	<PPI descriptive name>	PPI-001 Average time of RFC analysis
Process	<process ID the PPI is related to>	Request for change (RFC)
Goals	<strategic or operational goals the PPI is related to>	BG-002: Improve customer satisfaction BG-014: Reduce RFC time to response
Definition	The PPI is defined as { <DurationMeasure> <CountMeasure> <ConditionMeasure> <DataMeasure> <DerivedMeasure> <AggregatedMeasure> } [expressed in <unit of measure>]	The PPI is defined as the average of Duration of Analyse RFC activity
Target	The PPI value must { be { greater lower } than [or equal to] <bound> be between <lower bound> and <upper bound> [inclusive] fulfil the following constraint: <target constraint> }	The PPI value must be slower than or equal to 1 working day
Scope	The process instances considered for this PPI are { the last <n> ones those in the analysis period <AP-x> }	The process instances considered for this PPI are the last 100 ones
Source	<source from which the PPI measure can be obtained>	Event logs of BPMS
Responsible	{ <role> <department> <organization> <person> }	Planning and quality manager
Informed	{ <role> <department> <organization> <person> }	CIO
Comments	<additional comments about the PPI>	Most RFCs are created after 12:00

Friedenstab et al. argue that such linguistic patterns do not fit to the usually graphical modeling of processes which makes integration difficult [12]. The authors discuss some more approaches to BAM modeling. With regard to the limitations revealed, they present a BAM-related extension of the graphical Business Process Model & Notation (BPMN) [12].

Using an abstract language syntax based on the Unified Modeling Language (UML) they started defining meta models for language constructs needed for BAM as there are Duration, Frequency, Composited Basic Measure, Aggregated Measure, Filter, Target Definition, Actions, Measure-based Expressions and Dashboard. Figure 8 depicts the example for the duration of elements on different levels of detail, where the grey colored parts indicate references to the BPMN specification.

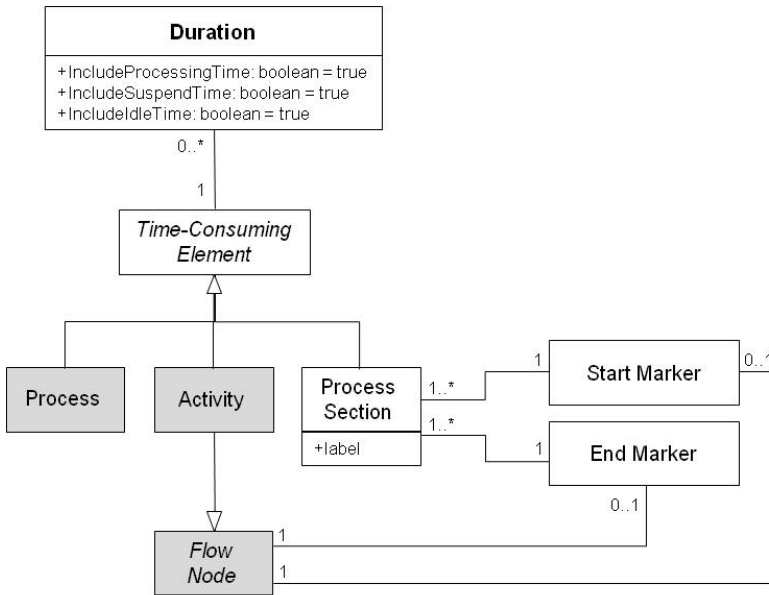


Fig. 8. Meta Model for Duration (related to BPMN) [12]

In a second step Friedenstab et al. developed a concrete syntax allowing for modeling the abstract language elements with graphical symbols and text labels. Parts of it are visible in figure 9. The example shows the BAM model for determining the cycle times of a purchase order process modeled in BPMN (lower part). The upmost part for example expresses the fact that the overall cycle time (Duration) for the last 50 instances (Filter) has to be determined and displayed on the dashboard (Dashboard). Monitoring the average of the overall cycle time for completed instances controls the modeled business logic of the process. If it is above 48 hours goods are delivered with an express shipping. Otherwise standard shipping is carried out. A deviation also leads to an alert sent to the process owner, while in any case the average is to be presented on the dashboard. The latter is also valid for the third time-related metric in the example, the partial cycle-time for the company-internal part of the process, which is set into relation with the overall cycle time.

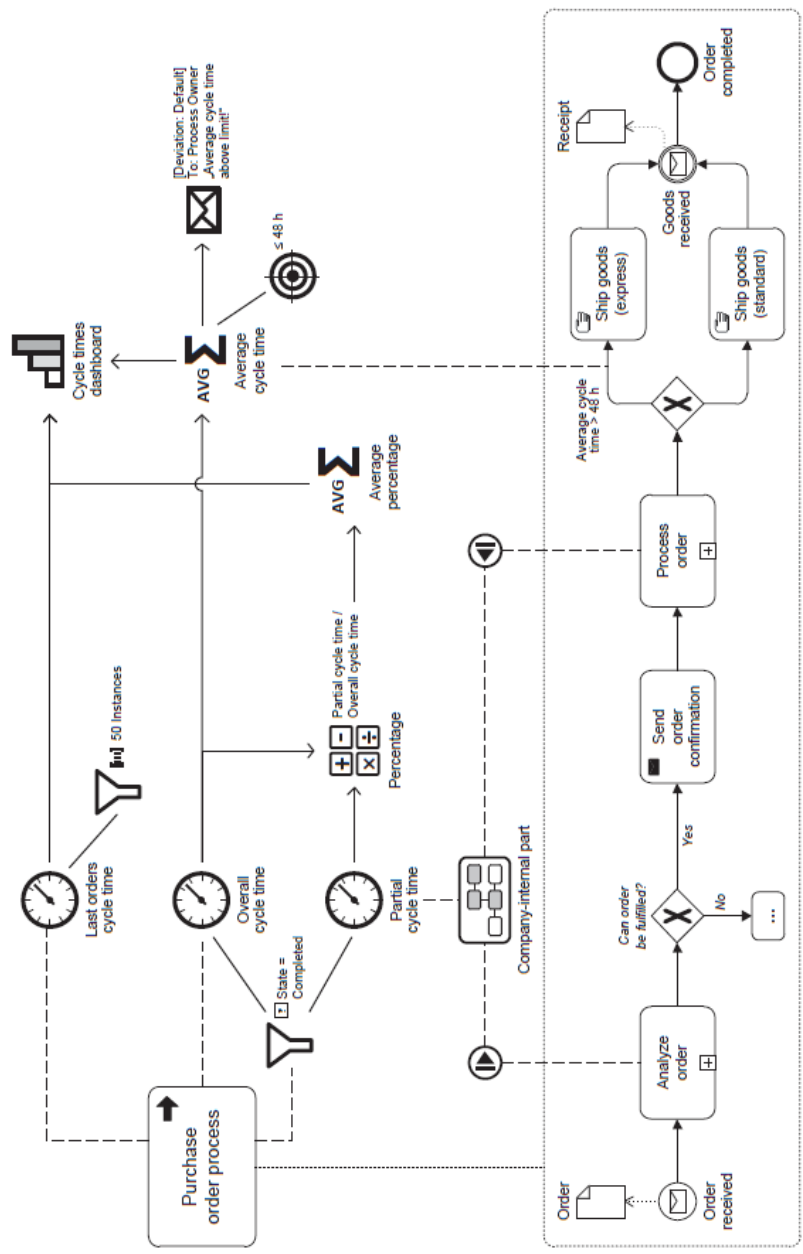


Fig. 9. BAM Model for Cycle Times of a Purchase Order Process based on BPMN [12]

The concept presented by Friedenstab et al. is thoroughly thought-out and clearly and precisely elaborated. The idea now is to adapt it to Subject-oriented Business Process Management and relate the abstract syntax to the S-BPM meta model instead of BPMN. Due to S-BPM being a more precise and comprehensive notation than BPMN [2] the mapping should be possible without problems. Table 3 compares the BPMN specification elements used by [12] with the ones appropriate in S-BPM [9].

Table 3. BPMN and S-BPM Specifications used in BAM Constructs

BAM Language Syntax Construct	Used BPMN Specification Element	Suitable S-BPM Specification Element
Duration (Time-Consuming Element)	Process, Activity, Flow Nodes	Process, Subject Behaviour States (Function, Send, Receive, Start, End)
Frequency (Countable Element)	Process, Activity, Data Objects, Data States	Process, Subject Behaviour States (Function, Send, Receive), Business Objects and their States
Actions	Process	Process
Measure-based Expressions	Expression, Sequence Flow	Incoming Message

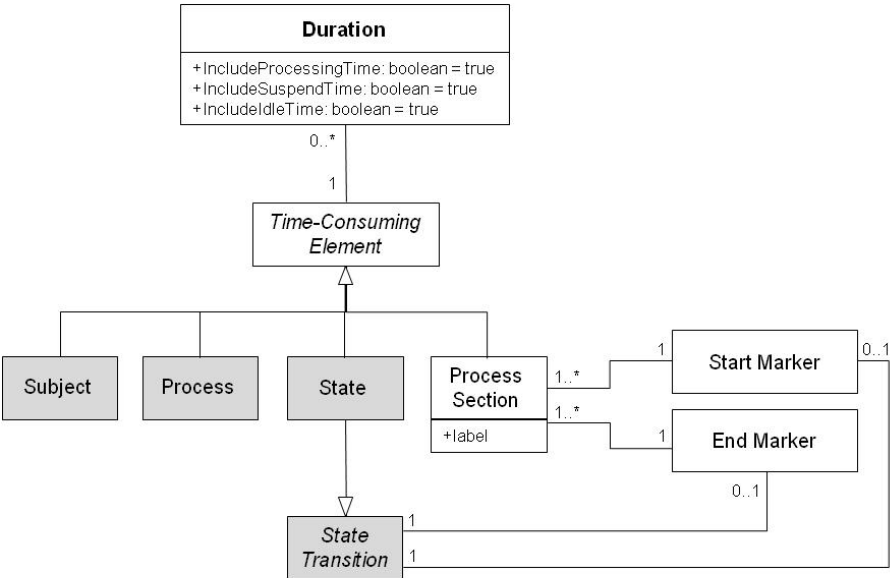


Fig. 10. Meta Model for Duration (related to S-BPM)

The remaining constructs as well as the extensions do not depend on the process modeling language and thus are not included in the table.

On the other hand S-BPM, following its paradigm of regarding subjects, predicates and objects as equally important parts of a process, offers the subject as an additional specification element to add¹. In figure 10 we modified the picture of figure 8 by replacing the BPMN by S-BPM elements and adding the subject. This allows modeling the determination of the overall time a subject (respectively the allocated resource(s)) spends on working on a process instance. This is of interest for cost-related analysis (see section 2.3).

In order to show how the BAM language syntax constructs can be related to subject-oriented models we designed the purchase order process in S-BPM. Due to missing information in the BPM model some assumptions were necessary like who performs the process steps (subjects). We then added the BAM modeling symbols to create a monitoring model similar to that in figure 9.

The result is depicted in the following graph. In the lower part it includes the subject interaction diagram (SID) of the process. The SID shows the subjects involved and how they coordinate themselves in the course of action by exchanging messages. In the monitoring model in the upper part a difference can be seen. The partial cycle time for the company-internal activities can be modeled by just relating the clock symbol to the subject 'Sales'. In the example this subject represents all steps carried out within the organization. In the same way we can determine the cycle time for the other subjects.

Given a special information demand a more granular modeling of BAM parameters is possible on the subject behavior level. Figure 12 for example details the behavior of 'Sales' including all receive, send and functional states walked through by the subject. The symbols indicate that the average cycle time between order reception and confirming the order to the customer should be measured. In the same way cycle times between states in behaviours of different subjects can be modeled.

Back on the level of subject interaction diagram we could also model to determine the overall time for receiving (waiting), sending and doing, both by process and by subject. Modeling on the two diagram levels reduces complexity.

4 Conclusion and Future Work

This contribution systematized Business Process Monitoring and shed some light on the current state of monitoring in the context of S-BPM. Starting there we emphasized Business Activity Monitoring and took a closer look to the modeling of BAM parameters. We showed that the approach for BPMN presented by Friedenstab et al. can be adapted to S-BPM with little effort and that S-BPM shows additional potential to further develop the concept.

¹ Including the BPMN pools as a BAM language syntax construct on the BPMN side could offer a comparable opportunity.

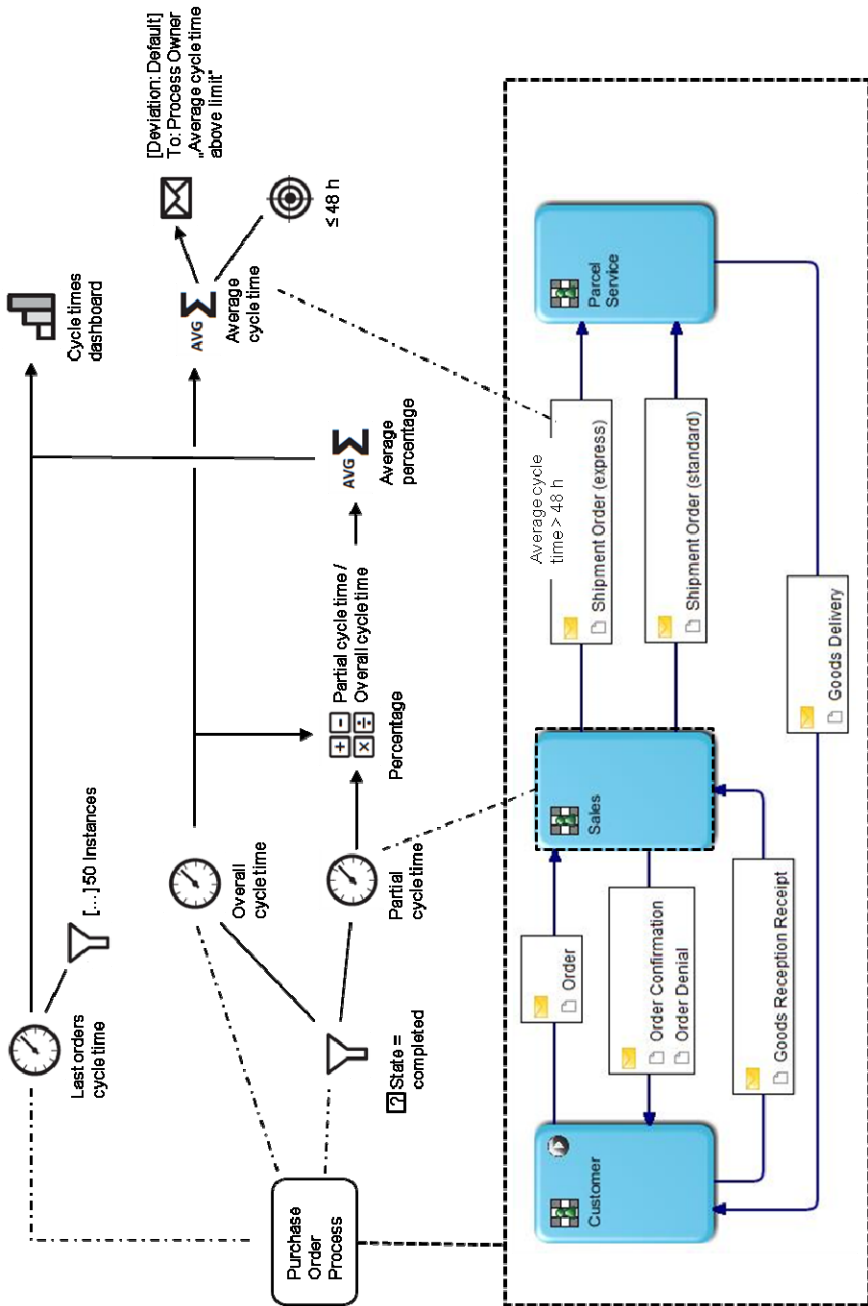


Fig. 11. BAM Model for Cycle Times of a Purchase Order Process based on S-BPM

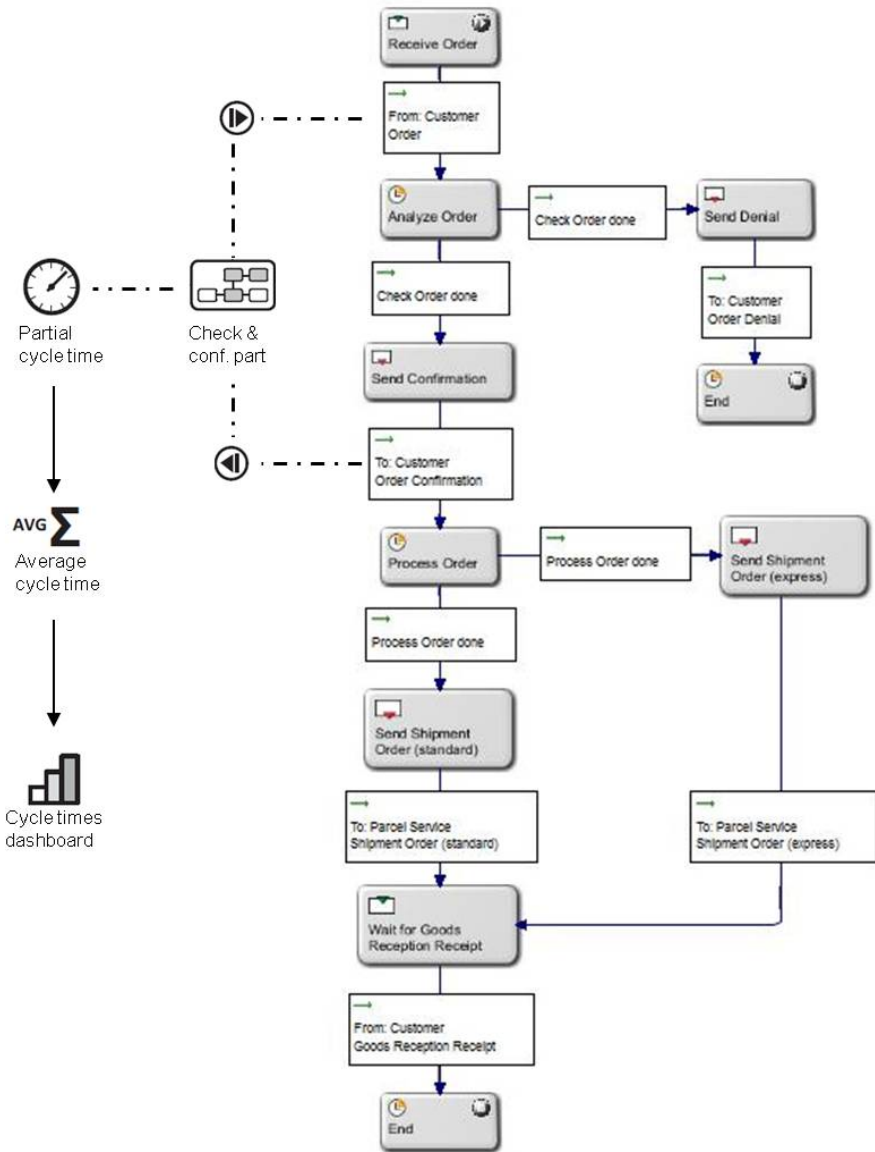


Fig. 12. BAM Model for Cycle Time of a Process Section based on S-BPM

This is where future work needs to continue. The first step to go should be elaborating, detailing and tailoring the modeling language for the needs of S-BPM in order to define a comprehensive basis for the subsequent task of developing easy-to-use software tool support. These could be pursued as parts of the Open S-BPM Initiative, which is underway to develop – among other artifacts - a process modeling

and a process execution software for S-BPM, both based on the Eclipse Ecore meta model (see [11, 14] and <http://www.i2pm.net/open-s-bpm/projects>). The BAM modeling tool should make use of the model data created by the process modeling tool and add its results to a combined model base both for process execution and BAM. The third step would then be to build BAM functionality which at runtime processes events according to the subject-oriented BAM models. There is some evidence that S-BPM provides good premises to support event processing [10], e.g. by integration of a CEP Engine or single Event Processing Agents as subjects into processes. This has to be proven by a thorough investigation of the interrelationship and integration potential of CEP and S-BPM concepts and solutions. In such an integrated environment a S-BPM-based process engine like Metasonic Flow would not only be event producer and event consumer, but also part of the event processing logic, communicating with other elements by message exchange (see figure 7). Generating code for BAM functionality out of the BAM model is another challenge to tackle [12]. With the clear formal semantic of its underlying process algebra the S-BPM modeling language allows this for process models [9]. Consequently it seems to be worthwhile to investigate the possibilities of code generation out of BAM models too.

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